

P-54 Action, Parameter, and Convention Card

A fail-closed definition of the four-dimensional Route-F baseline

Route-F four-dimensional mainline

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Abstract

This card removes an ambiguity hidden in the shorthand $54_H + 126_H + 10_{H,\mathbb{C}}$. With only those gauge representations and no extra symmetry, both 10_H and 10_H^* couple to the three chiral 16_F s; the Yukawa sector therefore contains three complex symmetric matrices. The predictive two-matrix model instead needs a global Peccei–Quinn symmetry and a complex gauge singlet S . We freeze that latter theory as P54PQ-v2; the no-PQ theory is retained as the comparison model P54N, not silently mixed with it.

The complete canonically normalized renormalizable action, tensor and charge conventions, scalar potential, vacuum coordinates, parameter quotient, and interfaces to the spectrum, flavor, and proton-decay calculations are given below. A direct charge audit finds that the often-quoted printed η_1 contraction in Eq. (24) of Babu–Khan is incompatible with the PQ charges printed in their Eq. (23). Charge conservation uniquely moves one complex conjugation. An independent representation-ring audit also finds the previously omitted invariant $\chi_7 \Phi_{ij} \phi_i \phi_j S^* + \text{h.c.}$. The corrected action has three rephasing-invariant scalar CP phases. These are definition-level corrections: a calculation that uses the printed monomial and simultaneously claims continuous PQ symmetry or omits χ_7 is not a calculation in P54PQ-v2.

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1 Status and branch identity

The primary P-layer action is

P54PQ-v2 : $\text{Spin}(10) \times U(1)_{\text{PQ}}, \quad 3(16_F) + 54_{H,\mathbb{R}} + 126_{H,\mathbb{C}} + 10_{H,\mathbb{C}} + 1_{H,\mathbb{C}}.$

(1)

The singlet is not cosmetic: after the 126_H expectation value, it removes the additional spontaneously broken global direction and supplies the axion sector. The gauge group is $\text{Spin}(10)$, not merely $SO(10)$, because the matter representation is a chiral half-spinor.

Two nearby theories are explicitly outside the frozen action:

$$\begin{aligned} \text{P54N} : & \quad 3(16_F) + 54_{\mathbb{R}} + 126_{\mathbb{C}} + 10_{\mathbb{C}}, \quad \text{no PQ and no singlet;} \\ \text{P210} : & \quad 210_H + 10_H + 120_H + \overline{126}_H, \quad \text{comparison/rescue only.} \end{aligned}$$

The first has the exact shorter field list but a different, three-matrix Yukawa action. The second has different vacuum and flavor freedom. Neither may lend parameters, thresholds, or Clebsch coefficients to P54PQ-v2.

2 Spacetime, generators, and fields

We use Minkowski signature $\eta_{\mu\nu} = \text{diag}(+, -, -, -)$, $\epsilon^{0123} = +1$, and two-component left-handed Weyl spinors. Long roots have length squared two. Hermitian generators obey

$$[T^A, T^B] = if^{ABC}T^C, \quad \text{tr}_{16}(T^AT^B) = 2\delta^{AB}. \quad (2)$$

The gauge field and curvature are

$$D_\mu = \partial_\mu - ig_{10}A_\mu^AT^A, \quad F_{\mu\nu}^A = \partial_\mu A_\nu^A - \partial_\nu A_\mu^A + g_{10}f^{ABC}A_\mu^BA_\nu^C. \quad (3)$$

Vector indices $i, j, \dots = 1, \dots, 10$ are contracted with δ_{ij} . The scalar tensors are

$$\Phi_{ij} = \Phi_{ji} = \Phi_{ij}^*, \quad \Phi_{ii} = 0, \quad \Phi \sim 54_{\mathbb{R}}, \quad (4)$$

$$\Sigma_{ijklm} = \Sigma_{[ijklm]}, \quad \star_{10}\Sigma = +i\Sigma, \quad \Sigma \sim 126_{\mathbb{C}}, \quad (5)$$

$$\phi_i \in \mathbb{C}^{10}, \quad \phi \sim 10_{\mathbb{C}}, \quad (6)$$

$$S \in \mathbb{C}, \quad S \sim 1_{\mathbb{C}}. \quad (7)$$

Interchanging $126 \leftrightarrow \overline{126}$ and the sign in (5) everywhere is a representation-label convention, not a second model. Our sign is fixed by the Yukawa contraction below.

The covariant derivatives in tensor notation are

$$(D_\mu\Phi)_{ij} = \partial_\mu\Phi_{ij} - ig_{10}A_\mu^A[(T_{10}^A)_i{}^k\Phi_{kj} + (T_{10}^A)_j{}^k\Phi_{ik}], \quad (8)$$

$$(D_\mu\phi)_i = \partial_\mu\phi_i - ig_{10}A_\mu^A(T_{10}^A)_i{}^j\phi_j, \quad (9)$$

$$(D_\mu\Sigma)_{i_1\dots i_5} = \partial_\mu\Sigma_{i_1\dots i_5} - ig_{10}A_\mu^A\sum_{r=1}^5(T_{10}^A)_{i_r}{}^j\Sigma_{i_1\dots i_{r-1}ji_{r+1}\dots i_5}, \quad (10)$$

$$D_\mu S = \partial_\mu S. \quad (11)$$

The continuous PQ charges are fixed, rather than defined only up to a sign, by

$$q_{\text{PQ}}(\Psi, \Phi, \Sigma, \phi, S) = (-1, 0, +2, -2, -4). \quad (12)$$

Here $\Psi_a \sim 16_F$, $a = 1, 2, 3$. Multiplying all charges by -1 is of course equivalent, but changing only one entry is not.

3 The complete renormalizable action

The Lagrangian is

$$\mathcal{L}_{\text{P54PQ}} = \mathcal{L}_{\text{gauge}} + \mathcal{L}_{\text{kin}} + \mathcal{L}_Y - V(\Phi, \Sigma, \phi, S), \quad (13)$$

with

$$\mathcal{L}_{\text{gauge}} = -\frac{1}{4}F_{\mu\nu}^A F^{A\mu\nu} + \frac{\theta_{10}g_{10}^2}{32\pi^2}F_{\mu\nu}^A \tilde{F}^{A\mu\nu}, \quad (14)$$

$$\begin{aligned} \mathcal{L}_{\text{kin}} = & i\Psi_a^\dagger \bar{\sigma}^\mu D_\mu \Psi_a + \frac{1}{2}(D_\mu \Phi_{ij})(D^\mu \Phi_{ij}) + \frac{1}{5!}(D_\mu \Sigma_{ijklm})^*(D^\mu \Sigma_{ijklm}) \\ & + (D_\mu \phi_i)^*(D^\mu \phi_i) + (\partial_\mu S)^*(\partial^\mu S), \end{aligned} \quad (15)$$

where $\tilde{F}^{\mu\nu} = \epsilon^{\mu\nu\rho\sigma} F_{\rho\sigma}/2$. Gauge fixing, Faddeev–Popov ghosts, and counterterms are introduced only after a gauge and renormalization scheme are declared; they are not new parameters of the gauge-invariant action.

3.1 Yukawa sector

Let C_{10} denote the charge-conjugation matrix on the spinor indices and $\Gamma_{i_1\dots i_5}$ the antisymmetrized Clifford product projected onto the self-duality selected in (5). Our normalization is

$$-\mathcal{L}_Y = \frac{1}{2}(Y_{10})_{ab} P s i_a^T C_{10} \Gamma_i \Psi_b \phi_i^* + \frac{1}{2 \cdot 5!}(Y_{126})_{ab} P s i_a^T C_{10} \Gamma_{ijklm} \Psi_b \Sigma_{ijklm} + \text{h.c.} \quad (16)$$

Both terms have total PQ charge zero. Because

$$16 \otimes 16 = 10_s \oplus 120_a \oplus 126_s, \quad (17)$$

$Y_{10} = Y_{10}^T$ and $Y_{126} = Y_{126}^T$. The coupling to ϕ , as opposed to ϕ^* , has PQ charge -4 and is absent. A 120_H and its antisymmetric Yukawa matrix are absent, not set to a small fitted value.

After Pati–Salam and electroweak breaking, define $h = Y_{10}$, $f = Y_{126}$ and the four bidoublet vacuum coordinates $v_{10}^{u,d}, v_{126}^{u,d}$. The Clebsch convention is

$$M_u = h v_{10}^u + f v_{126}^u, \quad M_d = h v_{10}^d + f v_{126}^d, \quad (18)$$

$$M_D = h v_{10}^u - 3f v_{126}^u, \quad M_e = h v_{10}^d - 3f v_{126}^d, \quad (19)$$

$$M_R = f \sigma, \quad M_\nu = M_L - M_D M_R^{-1} M_D^T. \quad (20)$$

Equation (20) keeps both type-I and type-II terms; setting $M_L = 0$ is a later fit hypothesis, not part of this action card.

3.2 Scalar potential

All repeated vector indices are summed. With the tensor normalizations in (4)–(5), the full PQ-invariant potential is

$$V = V_{\Phi\Sigma} + V_{\Phi\Sigma\phi} + V_S, \quad (21)$$

where

$$\begin{aligned} V_{\Phi\Sigma} = & -\frac{\mu^2}{2}\Phi_{ij}\Phi_{ij} + \frac{c}{3}\Phi_{ij}\Phi_{jk}\Phi_{ki} + \frac{a}{4}(\Phi_{ij}\Phi_{ij})^2 + \frac{b}{2}\Phi_{ij}\Phi_{jk}\Phi_{kl}\Phi_{li} - \frac{\nu^2}{2 \cdot 5!}\Sigma_{ijklm}\Sigma_{ijklm}^* \\ & + \frac{\lambda_0}{(2!)^2(5!)^2}(\Sigma_{ijklm}\Sigma_{ijklm}^*)(\Sigma_{nopqr}\Sigma_{nopqr}^*) \\ & + \frac{\lambda_2}{(4!)^2}\Sigma_{ijklm}\Sigma_{ijkln}^*\Sigma_{opqrm}\Sigma_{opqrn}^* \\ & + \frac{\lambda_4}{(3!)^2(2!)^2}\Sigma_{ijklm}\Sigma_{ijkno}^*\Sigma_{pqrlm}\Sigma_{pqrn}^* \\ & + \frac{\lambda'_4}{(3!)^2}\Sigma_{ijklm}\Sigma_{ijkno}^*\Sigma_{pqrlm}\Sigma_{pqrm}^* + \frac{\alpha}{2 \cdot 5!}\Phi_{ij}\Phi_{ij}\Sigma_{pqrlm}\Sigma_{pqrlm}^* \\ & + \frac{\beta}{3!}\Phi_{ij}\Phi_{kl}\Sigma_{mnoik}\Sigma_{mnojl}^*, \end{aligned} \quad (22)$$

$$\begin{aligned}
V_{\Phi\Sigma\phi} = & -\xi_0^2\phi_i\phi_i^* + \xi_1(\phi_i\phi_i^*)(\phi_j\phi_j^*) + \xi_2(\phi_i\phi_i^*)(\phi_j^*\phi_j^*) + \xi_3\Phi_{ij}\phi_i\phi_j^* \\
& + \frac{\gamma_1}{4!}\Sigma_{ijklm}\Sigma_{ijkln}^*\phi_m\phi_n^* + \frac{\gamma_2}{4!}\Sigma_{ijklm}\Sigma_{ijkln}^*\phi_n\phi_m^* + \frac{\eta_0}{2}\Phi_{ij}\Phi_{ij}\phi_k\phi_k^* \\
& + \frac{\eta_1}{(3!)^2(2!)^2}\underbrace{\Sigma_{ijklm}\Sigma_{ijkpq}^*\Sigma_{lmpqn}\phi_n}_{q_{\text{PQ}}=2-2+2-2=0} + \frac{\eta_1^*}{(3!)^2(2!)^2}\Sigma_{ijklm}^*\Sigma_{ijkpq}\Sigma_{lmpqn}^*\phi_n^* \\
& + \eta_2\Phi_{ij}\Phi_{ik}\phi_j\phi_k^* + \frac{\eta_3}{4!}\Sigma_{ijklm}\Sigma_{ijkln}\phi_m\phi_n + \frac{\eta_3^*}{4!}\Sigma_{ijklm}^*\Sigma_{ijkln}^*\phi_m^*\phi_n^*, \tag{23}
\end{aligned}$$

and

$$\begin{aligned}
V_S = & -\mu_s^2SS^* + \chi_1(SS^*)^2 + \chi_2\Sigma_{ijklm}\Sigma_{ijklm}^*SS^* + \chi_3\Phi_{ij}\Phi_{ij}SS^* \\
& + \frac{\chi_4}{4!}\Sigma_{ijklm}\Sigma_{ijkln}\Phi_{mn}S + \frac{\chi_4^*}{4!}\Sigma_{ijklm}^*\Sigma_{ijkln}^*\Phi_{mn}S^* + \chi_5\phi_i\phi_i^*SS^* \\
& + \chi_6\phi_i\phi_iS^* + \chi_6^*\phi_i^*\phi_i^*S + \chi_7\Phi_{ij}\phi_i\phi_jS^* + \chi_7^*\Phi_{ij}\phi_i^*\phi_j^*S. \tag{24}
\end{aligned}$$

The Φ_{mn} contraction in the χ_4 term is forced by its two free five-form indices. The often-reproduced Φ_{ij} version repeats i, j more than twice and is not valid Einstein notation; this card uses the index-balanced contraction. Couplings $\eta_1, \eta_3, \chi_4, \chi_6, \chi_7$ are complex; all other displayed scalar couplings are real. Reality of the action follows from the explicit Hermitian pairs.

The χ_7 term is forced by the unique 54 channel in $\text{Sym}^2(10) = 1 \oplus 54$: its charge is $q(\Phi) + 2q(\phi) + q(S^*) = 0 - 4 + 4 = 0$. It was absent from the published-style list used for version 1, so the independent basis audit changes the frozen model rather than treating the omission as a numerical uncertainty.

3.3 The corrected etal convention

The printed expression in Ref. [1] contains $\Sigma\Sigma^*\Sigma^*\phi$. With its printed charges, that monomial has charge -4 and cannot occur in a continuously PQ-invariant action. For a quartic with three Σ -type fields and one ϕ -type field, charge neutrality requires

$$2(n_\Sigma - n_{\Sigma^*}) - 2(n_\phi - n_{\phi^*}) = 0. \tag{25}$$

The monomial inducing the 126 bidoublet expectation value and paired with its Hermitian conjugate is therefore uniquely $\Sigma\Sigma^*\Sigma\phi + \text{h.c.}$, which is the term in (23). We interpret the extra star in the printed formula as a conjugation typo. The no-PQ branch may use the printed contraction, but then it is not P54PQ-v2.

There is a second, purely index-level correction in the same printed potential. In $\Sigma_{ijklm}\Sigma_{ijkln}\Phi S$, the unpaired indices are m, n , so the only displayed contraction is Φ_{mn} , as in (24). Writing Φ_{ij} would make i, j occur three times and leave m, n free. This correction changes neither PQ charge nor parameter count.

4 Parameter card and quotient

Every scalar has engineering dimension one. The coefficient card is

Class	Parameters	Raw reals
Gauge/topological	g_{10}, θ_{10}	2
Scalar, dimension 2	$\mu^2, \nu^2, \xi_0^2, \mu_s^2$	4
Scalar, dimension 1	c, ξ_3 real; χ_6 complex	4
Scalar, dimension 0, real	$a, b, \lambda_0, \lambda_2, \lambda_4, \lambda'_4, \alpha, \beta, \xi_1, \xi_2, \gamma_1, \gamma_2, \eta_0, \eta_2, \chi_1, \chi_2, \chi_3, \chi_5$	18
Scalar, dimension 0, complex	$\eta_1, \eta_3, \chi_4, \chi_7$	8
Yukawa	two complex symmetric 3×3 matrices	24
Total	before field-basis quotient	60

The phase-charge vectors of the five complex scalar couplings are computed under the field redefinition

$$(\Sigma, \phi, S) \mapsto (e^{i\alpha_\Sigma} \Sigma, e^{i\alpha_\phi} \phi, e^{i\alpha_S} S). \quad (26)$$

They are

$$v_{\eta_1} = (1, 1, 0), \quad v_{\eta_3} = (2, 2, 0), \quad v_{\chi_4} = (2, 0, 1), \quad v_{\chi_6} = (0, 2, -1), \quad v_{\chi_7} = (0, 2, -1). \quad (27)$$

Their rank is two and their common null vector is $(1, -1, -2)$, the PQ direction in (12). Hence only two coupling phases can be removed. We choose

$$\eta_1 \geq 0, \quad \chi_4 \geq 0, \quad \begin{aligned} \delta_1 &= \arg \eta_3 - 2 \arg \eta_1, & \delta_2 &= \arg \chi_4 + \arg \chi_6 - 2 \arg \eta_1, \\ \delta_3 &= \arg \chi_7 - \arg \chi_6. \end{aligned} \quad (28)$$

There are therefore $34 - 2 = 32$ scalar parameters after scalar-field rephasing, including three physical scalar CP phases.

The two Yukawa matrices contain 24 real numbers. A generic $U(3)$ family basis removes nine; equivalently one may Takagi-diagonalize

$$Y_{10} = \text{diag}(y_1, y_2, y_3), \quad y_i \geq 0, \quad (29)$$

and leave Y_{126} complex symmetric. This gives 15 physical Yukawa parameters. The classical parameter space is thus

$$\dim_{\mathbb{R}}(\Theta_{\text{P54}}/\mathcal{G}_{\text{basis}}) = 2 + 32 + 15 = 49. \quad (30)$$

At the quantum level the anomalous PQ redefinition trades θ_{10} for the axion origin, so there are 48 independent continuous parameters in zero-temperature observables, plus discrete anomaly/domain-wall data. This last reduction is valid only if PQ quality is not spoiled by higher-dimension operators.

For contrast, P54N has three complex symmetric Yukawa matrices, 36 raw real Yukawa entries and $36 - 9 = 27$ after the generic family quotient. It is not a two-Yukawa-matrix realization of the shorter field list.

5 Vacuum and breaking conventions

The intended chain, including the global symmetry bookkeeping, is

$$\text{Spin}(10) \times U(1)_{\text{PQ}} \xrightarrow{\langle \Phi \rangle} G_{\text{PS}} \rtimes D \times U(1)_{\text{PQ}} \xrightarrow{\langle \Sigma \rangle} G_{\text{SM}} \times U(1)'_{\text{PQ}} \xrightarrow{\langle S \rangle} G_{\text{SM}} \xrightarrow{\langle h \rangle} SU(3)_C \times U(1)_{\text{em}}, \quad (31)$$

where $G_{\text{PS}} = (SU(4)_C \times SU(2)_L \times SU(2)_R)/\mathbb{Z}_2$ in the global form inherited from $\text{Spin}(10)$. The 54 expectation value is

$$\langle \Phi \rangle = \omega_s \text{diag} \left(-\frac{2}{5}, -\frac{2}{5}, -\frac{2}{5}, -\frac{2}{5}, -\frac{2}{5}, -\frac{2}{5}, \frac{3}{5}, \frac{3}{5}, \frac{3}{5}, \frac{3}{5} \right). \quad (32)$$

It is traceless, preserves $SO(6) \times SO(4) \simeq G_{\text{PS}}$, and is D even. Our remaining radial coordinates are

$$\langle \Sigma_{246810} \rangle = \frac{\sigma}{4\sqrt{2}}, \quad \langle S \rangle = \frac{v_s}{\sqrt{2}}, \quad \langle h^0 \rangle = \frac{v}{\sqrt{2}}, \quad (33)$$

with $\omega_s, \sigma, v_s, v \geq 0$ after the stated phase conventions.

The relevant Pati–Salam decompositions are

$$10 = (6, 1, 1) \oplus (1, 2, 2), \quad (34)$$

$$54 = (1, 3, 3) \oplus (6, 2, 2) \oplus (20', 1, 1) \oplus (1, 1, 1), \quad (35)$$

$$126 = (6, 1, 1) \oplus (10, 3, 1) \oplus (\overline{10}, 1, 3) \oplus (15, 2, 2). \quad (36)$$

The neutral member of $(\overline{10}, 1, 3)$ carries σ . The 54 keeps D parity, so $g_L = g_R$ at the Pati–Salam matching scale in this branch.

Substituting (32) into its self-potential gives the exact check

$$V_{54}(\omega_s) = -\frac{6}{5}\mu^2\omega_s^2 + \frac{4}{25}c\omega_s^3 + \frac{36}{25}a\omega_s^4 + \frac{42}{125}b\omega_s^4. \quad (37)$$

Indeed $\text{tr } X^2 = 12/5$, $\text{tr } X^3 = 12/25$, and $\text{tr } X^4 = 84/125$ for the diagonal matrix X in (32). These rational coefficients are convention fingerprints: a spectrum code with different values is not using this card.

The electroweak doublet is one normalized zero mode of the four-doublet system

$$h_{\text{SM}} = c_1 H_u + c_2 \epsilon H_d^* + c_3 \Phi_u + c_4 \epsilon \Phi_d^*, \quad \sum_{i=1}^4 |c_i|^2 = 1. \quad (38)$$

The H doublets are from $(1, 2, 2)_{10}$ and the Φ doublets here are from $(15, 2, 2)_{126}$; this reuse of Φ is common in Pati–Salam papers but must not be confused with the $SO(10)$ tensor Φ_{ij} in this card. In code they shall be named `H10u`, `H10d`, `H126u`, `H126d`. The required fine tuning is exactly one small eigenvalue of the complete doublet mass matrix. The color-triplet matrix must remain nonsingular and is never tuned by hand independently.

6 Observable map and P1 interface

The action defines, but this card does not yet numerically evaluate, the map

$$\mathcal{F} : \Theta_{\text{P54}}/\mathcal{G}_{\text{basis}} \longrightarrow \mathcal{O}, \quad (39)$$

where $\mathcal{O}$ contains the full scalar and vector spectrum, matched gauge couplings, fermion singular values and mixing matrices, neutrino observables, proton partial widths, and axion observables. The future Jacobian is

$$J_{Ai} = \frac{\partial \mathcal{O}_A}{\partial \vartheta_i}, \quad \text{rank } J \leq \min(N_{\mathcal{O}}, 48) \quad (40)$$

after the quantum PQ quotient. No numerical rank is claimed before P1 exports a differentiable spectrum and P2–P4 fix matching conventions. The rank must be reported from a scaled SVD with singular-value tolerance and uncertainty propagation, not inferred from the number of fitted data.

The P1 spectrum exporter must consume exactly

$$\{g_{10}; \mu^2, \nu^2, \xi_0^2, \mu_s^2; c, \xi_3, \chi_6; \text{ all dimensionless scalar couplings including } \chi_7; Y_{10}, Y_{126}\} \quad (41)$$

in the conventions above and return, for every physical state,

$$\{G_{\text{SM}} \text{ irrep}, D \text{ parity}, m^2, \text{mixing vector}, \partial m^2 / \partial \vartheta_i, \text{Goldstone flag}\}. \quad (42)$$

Acceptance requires: all stationarity equations; equality between broken generator directions and scalar zero modes; positive physical Hessian at the candidate vacuum; the complete doublet and

triplet matrices; and comparison against competing stationary points. “Extended survival” may be a scan prior but may not replace this returned spectrum.

The Wilsonian action is taken at a matching scale μ_0 below a declared cutoff Λ_{P54} . The minimal consistency contract is

$$M_U < \Lambda_{\text{P54}} \leq M_{\text{Pl}}, \quad \max\{|\lambda_i|, |Y_{ij}|^2, g_{10}^2\} < 4\pi, \quad (43)$$

supplemented in P2 by running perturbative-unitarity eigenvalues. Planck-suppressed PQ-violating operators are an explicit PQ-quality debt, not part of the renormalizable fit.

7 Freeze rules and remaining debts

The action card is frozen under the following rules.

1. The corrected η_1 term in (23) and the χ_7 term in (24) are authoritative. Changing either requires a new version and a new charge audit.
2. No 120_H , 210_H , extra 10_H , generic M_R , or arbitrary threshold mass may be imported into the primary branch.
3. The singlet S and PQ anomaly data are part of P54PQ-v2. Dropping S defines another theory.
4. Doublet fractions are derived eigenvectors of one scalar mass matrix; they are not independent fit parameters in the final parameter count.
5. M_I , M_U , and every heavy mass are derived from the vacuum and couplings. They may be scan coordinates only with the corresponding stationarity equations imposed.

Version 2 is the result of the independent invariant-basis audit; its proof and the PQ anomaly/global-form audit are recorded in the companion audit. The machine-generated P1 Hessian/spectrum is now recorded in its own companion and closes the tree-level algebraic vacuum gate. The remaining card-level debt is PQ quality above Λ_{P54} ; running, matching, flavor, and proton tests remain outside this card. The label is therefore “tree-level candidate”, not “phenomenologically viable theory”.

References

- [1] K. S. Babu and S. Khan, “A Minimal Non-Supersymmetric SO(10) Model: Gauge Coupling Unification, Proton Decay and Fermion masses,” [arXiv:1507.06712](#) (2015).
- [2] N. Haba, Y. Shimizu, and T. Yamada, “Neutrino Mass in Non-Supersymmetric SO(10) GUT,” [arXiv:2304.06263](#), Phys. Rev. D 108, 095005 (2023).